ZS-F7 v1.0

Reuleaux Geometry of the Z-Sector Boundary

Variational Derivation, J-Compatibility, and the Blaschke–Lebesgue Correspondence

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Verification: 18/18 PASS | Zero Free Parameters

§0. Abstract

We derive the cross-sectional geometry of the Z-sector mediation boundary from the Z-Spin scalar-tensor action with zero free parameters. The derivation proceeds through a six-step chain: (1) dim(Z) = 2 (PROVEN, ZS-F5) establishes the cross-section as a plane curve; (2) U(1)_Z symmetry of the action (PROVEN, ZS-F1) requires the cross-section to be a curve of constant width via the J-compatibility condition ε_J = 0 ⇔ h(θ) + h(θ+π) = const (ZS-M7 Theorem 4, PROVEN); (3) the frozen ε-field at the late-time attractor ρ = 1 (DERIVED, ZS-F3) reduces the shape determination to a 1-loop variational problem; (4) the Seeley–DeWitt expansion of the spectral zeta function establishes ∂Γ_Z/∂(Area) > 0 (DERIVED); (5) convexity of the cross-section; (6) the Blaschke–Lebesgue theorem (PROVEN, 1915) uniquely selects the Reuleaux triangle as the minimum-area convex constant-width curve.

The non-circularity amplitude $a_3/(w/2) = 1/8$ is the geometric extremum $1/(n^2 - 1)$ at n = 3, where n is the Z-sector polygon number (Face-Polygon Correspondence, PROVEN). We prove the Perturbative No-Go Theorem: the C_3-summed Coleman–Weinberg effective potential contains no cos(3θ) harmonic at any finite perturbative order (only cos(6mθ) survives C_3 projection), confirming the Reuleaux geometry as a variational extremum. The Single-Polyhedron U(1) Exactness Theorem (PROVEN by explicit computation on the truncated octahedron E_g eigenspace, θ-variation < 10^{-15}) establishes that U(1) breaking arises exclusively from the inter-sector O_h/I_h frame mismatch.

Three Seeley–DeWitt coefficients characterize the Reuleaux cross-section: $a_0 = (\pi - \sqrt{3})w^2/(8\pi)$, a_{1/2} = $-\pi w/(4\sqrt{\pi})$ (Barbier: identical to circle), and $a_1 = 1/2$ (corner contribution $\Delta a_1 = 1/3 = 1/X$ from three vertices at angle π/X). The functional equation D_ζ(s) = D_ζ(1−s) is preserved at all orders. Anti-numerology Monte Carlo (500,000 trials): P(random match for area deficit) < 0.8%. Verification: 18/18 PASS.

Keywords: Reuleaux triangle, Z-sector geometry, Blaschke–Lebesgue theorem, constant-width curve, variational principle, Seeley–DeWitt expansion, J-symmetry, support function, polyhedral cross-section, spectral zeta function

Epistemic Status Legend

Status	Definition
PROVEN	Exact mathematical fact, verified to machine precision
DERIVED	Follows from ZS axioms with complete chain; no free parameters
DERIVED-CONDITIONAL	Derived under an explicitly stated assumption
VERIFIED	Numerical computation confirms analytical claim to stated precision
TESTABLE	Quantitative prediction with pre-registered falsification condition
HYPOTHESIS	Motivated by framework, requires further verification
OBSERVATION	Empirically validated but theoretical derivation pending
NON-CLAIM	Explicitly disclaimed. This paper does NOT assert this claim
OPEN	Recognized gap requiring future work
LOCKED	Input imported from upstream paper. Not re-derived here

§1. Introduction and Scope

1.1 What This Paper Answers

The Z-sector of Z-Spin Cosmology mediates all information transfer between the X-sector (space-particle, dim = 3) and Y-sector (time-wave, dim = 6) through a 2-dimensional boundary surface. The block-Laplacian identity L_XY ≡ 0 (PROVEN, ZS-F1) forces every X↔Y transition through the Z-mediator, establishing the Z-sector as the universal bottleneck of the framework. Prior papers characterize the Z-sector algebraically (ZS-F5: dim(Z) = 2 from the j = 1/2 recoupling uniqueness theorem, PROVEN), dynamically (ZS-F1: the Z-bias field Φ = ρe^{iθ} with double-well potential, PROVEN), and polyhedrally (ZS-F2: tetrahedron pair/stella octangula mediating between truncated octahedron and truncated icosahedron, PROVEN).

One question has remained open: what is the effective cross-sectional geometry of the Z-sector mediation boundary? The cross-section determines the spectral properties of the Z-mediator (through the Seeley–DeWitt heat kernel coefficients), constrains the Berry–Keating completion factor B(s) (ZS-QS §4.3, OPEN), and governs how curvature information is transmitted between the X and Y sectors. This paper derives the answer from the Z-Spin action: the cross-section is the Reuleaux triangle, uniquely selected by a variational principle isomorphic to the Blaschke–Lebesgue theorem of convex geometry.

The derivation uses no free parameters beyond those already locked in the Z-Spin framework. The Reuleaux triangle emerges not as an assumption but as the unique extremum of the 1-loop effective action under symmetry constraints inherited from the action itself. This result completes the foundational characterization of the Z-sector, adding geometric cross-section to the previously established algebraic, dynamical, and polyhedral properties.

1.2 Locked Inputs

Input	Value	Source	Status
A = 35/437	0.080092	ZS-F2 v1.0	LOCKED
(Z, X, Y) = (2, 3, 6), Q = 11	Slot register	ZS-F5 v1.0	PROVEN
U(1)_Z symmetry	$\Phi \to e^{i\alpha}\Phi$	ZS-F1 v1.0 §3.2	PROVEN
`L_XY ≡ 0`	Z-mediation forced	ZS-F1 v1.0	PROVEN
ε frozen at ρ = 1	$m_\varepsilon \approx 1.34\, M_P$	ZS-F3 v1.0	DERIVED
J: $\lvert j\rangle \to \lvert Q-1-j\rangle$	Z_2 seam involution	ZS-F5 v1.0	PROVEN
$\varepsilon_J(\sigma,t) = 0$ iff $\sigma = 1/2$	Unique J-intertwining	ZS-M7 v1.0 Thm 4	PROVEN
n = 3 (Z-sector polygon)	Face-Polygon Corr.	The Book §4.5	PROVEN
$\delta\varphi = A$ per cycle	Regge holonomy phase	ZS-M3 v1.0	DERIVED
E_g eigenvalue (TO)	$\lambda = 3 - \sqrt{3} \approx 1.268$	ZS-M6 v1.0	PROVEN
Frame mismatch angle	$\alpha = \pi/10 = 18°$	ZS-S6 v1.0 §3.3	DERIVED

1.3 Dependencies

Depends on: ZS-F1 (action, U(1)_Z, L_XY = 0), ZS-F2 (A = 35/437, polyhedral selection rules), ZS-F3 (ε-field attractor dynamics, mass spectrum), ZS-F5 (dim(Z) = 2, Q = 11, J-involution), ZS-M3 (Regge holonomy, δφ = A), ZS-M6 (heat kernel, block-Laplacian verification), ZS-M7 (transfer operator, ε_J theorem), ZS-S6 (O_h/I_h frame mismatch, non-abelian holonomy).

Downstream: ZS-M4/M7 (spectral bridge, Seeley–DeWitt input for B(s)), ZS-QS (Inverse Riemann Engine, seam geometry for P2 closure), ZS-S4 (Higgs VEV, Coleman–Weinberg mechanism), ZS-Q6 (Kelvin cell entanglement, boundary geometry).

§2. Mathematical Preliminaries: Curves of Constant Width

2.1 Support Function and Constant-Width Condition

A convex body K in the plane is uniquely characterized by its support function $h(\theta)$, defined as the signed distance from the origin to the tangent line with outward normal direction θ. The boundary curve is recovered via the parametric equations $x(\theta) = h\cos\theta - h'\sin\theta$ and $y(\theta) = h\sin\theta + h'\cos\theta$. The radius of curvature at angle θ is $\rho(\theta) = h(\theta) + h''(\theta)$, and convexity requires $\rho(\theta) \geq 0$ everywhere. For a convex body of constant width w, the width measured in any direction θ equals w:

$$h(\theta) + h(\theta + \pi) = w \quad \text{for all } \theta \in [0, 2\pi) \qquad (1)$$

Writing $h(\theta) = w/2 + f(\theta)$, the constant-width condition becomes $f(\theta) + f(\theta + \pi) = 0$, i.e., f is anti-periodic with period π. The Fourier expansion of f therefore contains only odd harmonics:

$$f(\theta) = \Sigma_{n=3,5,7,...} (a_n \cos n\theta + b_n \sin n\theta) \qquad (2)$$

The $n = 1$ harmonic is absent because it corresponds to a translation of the origin, not a shape change. The enclosed area is computed via the standard support-function formula $A = \frac{1}{2}\int[h^2 - (h')^2]d\theta$. Substituting $h = w/2 + f$ and using the orthogonality of Fourier modes:

$$\text{Area} = \pi w^2/4 - (\pi/2) \Sigma_{n \text{ odd} \geq 3} (n^2 - 1)(a_n^2 + b_n^2) \qquad (3)$$

Since $n^2 - 1 > 0$ for all $n \geq 3$, every non-zero Fourier coefficient strictly decreases the area below the circle value $\pi w^2/4$. The coefficient $(n^2 - 1)$ grows quadratically: for $n = 3$ it is 8, for $n = 5$ it is 24, for $n = 7$ it is 48. Higher harmonics are therefore more "efficient" at reducing area per unit amplitude, but their convexity bounds are correspondingly tighter. [STATUS: PROVEN, standard convex geometry, cf. Bonnesen-Fenchel 1934.]

2.2 Barbier's Theorem

Theorem 2.1 (Barbier, 1860). Every convex curve of constant width w has perimeter $L = \pi w$, independent of its shape.

Proof: The perimeter of a convex curve with support function h is $L = \int_0^{2\pi} h(\theta)\, d\theta$. Splitting the integral into $[0,\pi)$ and $[\pi,2\pi)$ and using $h(\theta+\pi) = w - h(\theta)$: $L = \int_0^{\pi} h\, d\theta + \int_0^{\pi} (w-h)\, d\theta = \pi w$. □

Physical significance: Barbier's theorem has a profound consequence for the Z-Spin spectral theory. The Seeley–DeWitt coefficient $a_{1/2}$, which depends on the boundary perimeter, is IDENTICAL for all constant-width curves with the same width w. This means the leading spectral correction is shape-independent within the class of constant-width curves, protecting the functional equation from shape-dependent perturbations. The perimeter is "topologically protected" by the constant-width constraint. [STATUS: PROVEN]

2.3 Blaschke–Lebesgue Theorem

Theorem 2.2 (Blaschke 1915, Lebesgue 1914). Among all convex curves of constant width w, the Reuleaux triangle uniquely achieves the minimum enclosed area:

$$\text{Area_min} = (\pi - \sqrt{3})w^2/2 \approx 0.70477\, w^2 \qquad (4)$$

For comparison, the circle of the same width has area $\pi w^2/4 \approx 0.78540\, w^2$. The Reuleaux triangle's area is 89.73% of the circle's.

The Reuleaux triangle is constructed from an equilateral triangle of side w by replacing each side with a circular arc of radius w centered at the opposite vertex. It has three arcs (each subtending $\pi/3$) and three

sharp vertices (each with interior angle $\pi/3$ where the radius of curvature $\rho = 0$). The dominant Fourier structure of its support function is:

$$h(\theta) = w/2 + (w/16) \cos 3\theta + \text{higher } C_3\text{-harmonics} \qquad (5)$$

The full Reuleaux support function involves all harmonics $n = 3, 9, 15, \ldots$ (C_3-symmetric odd harmonics), with amplitudes determined by the requirement that $\rho(\theta) = 0$ at exactly three points and $\rho(\theta) = w$ on the three arcs. The $n = 3$ truncation gives area $0.73631\ w^2$ (vs. exact $0.70477\ w^2$); the difference $0.03154\ w^2$ is contributed by harmonics $n \geq 9$. [STATUS: PROVEN]

2.4 Curvature Distribution: The 50/50 Split

The total turning angle of any convex closed curve is 2π (Umlaufsatz). For the Reuleaux triangle, this budget splits exactly in half between smooth and singular contributions:

Curvature Source	Number	Each Contribution	Total	Fraction
Smooth arcs ($\kappa = 1/w$)	3	$\pi/3$	π	1/2
Sharp vertices ($\rho = 0$)	3	$\pi/3$	π	1/2
Combined	6	—	2π	1

This 50/50 partition is unique among C_3-symmetric constant-width curves: any smoothing of the vertices shifts curvature from the discrete to the continuous channel, breaking the equal partition. Physically, the Z-sector mediates between continuous Y-sector dynamics (waves/time) and discrete X-sector geometry (particles/space). The equal curvature partition reflects this mediating role: exactly half the geometric information is processed through each channel. [STATUS: PROVEN for the curvature partition; physical interpretation is HYPOTHESIS.]

§3. Single-Polyhedron U(1) Exactness Theorem

3.1 Statement

Theorem 3.1 (Single-Polyhedron U(1) Exactness). On any Archimedean polyhedron P with symmetry group Γ, the quadratic form $Q(\theta) = \langle\Phi(\theta)|L|\Phi(\theta)\rangle$, where L is the graph Laplacian and $\Phi(\theta) = \cos\theta\ \varphi_1 + \sin\theta\ \varphi_2$ ranges over the E_g eigenspace, is exactly θ-independent.

Proof: The graph Laplacian $L = D - A$ (degree matrix minus adjacency) commutes with all symmetry operations: $[L, R_g] = 0$ for $g \in \Gamma$. The E_g eigenspace is an irreducible 2-dimensional representation of Γ, with basis $\{\varphi_1, \varphi_2\}$ satisfying $L\varphi_i = \lambda_{E_g}\varphi_i$. By Schur's lemma, any Γ-equivariant operator acting on an irreducible representation is proportional to the identity. Since $\langle\Phi|L|\Phi\rangle = \lambda_{E_g}\langle\Phi|\Phi\rangle = \lambda_{E_g}$ for normalized Φ, the quadratic form is constant on the unit circle in E_g. This extends to any Γ-invariant functional of Φ that is quadratic: vertex coupling Σ_v

$|\Phi_v|^2$, edge kinetic coupling $\Sigma_{ij} |\Phi_i - \Phi_j|^2$, and face-resolved projections are all θ-independent by the same argument. □ [STATUS: PROVEN]

3.2 Numerical Verification on the Truncated Octahedron

The truncated octahedron was explicitly constructed with 24 vertices at all permutations of (0, ±1, ±2), 36 edges at Euclidean distance √2, and 14 faces (8 hexagons + 6 squares). The complete eigenvalue spectrum of the graph Laplacian:

λ	Degeneracy	O_h Irrep	Role
0.0000	1	A_1g	Zero mode
0.5858	3	T_1u	—
1.2679	2	E_g	★ C_XZ source (Z-sector)
2.0000	3	T_2g	—
2.5858	3	T_1g	—
3.4142	3	T_2u	—
4.0000	3	A_2u+T_1u	—
4.7321	2	E_u	Second 2D
5.4142	3	T_2g	—
6.0000	1	A_1g	Maximum

The E_g eigenspace at $\lambda_{E_g} = 3 - \sqrt{3} \approx 1.26795$ (exact algebraic value) provides the 2D Z-sector coupling space (C_XZ matrix of ZS-S1 §3). Orthonormality was verified: $\langle\varphi_1|\varphi_2\rangle = 1.16 \times 10^{-16}$ (machine precision). The edges were classified as SH (square-hexagon boundary, 24 edges) and HH (hexagon-hexagon, 12 edges), and the θ-dependent coupling was computed at 3,601 uniformly spaced angles over [0, 2π):

Coupling Level	θ-Variation	Exact Value	Status
Vertex coupling f_SQ(θ)	$< 10^{-15}$	1.0000 (all θ)	EXACTLY ZERO
SH edge kinetic k_SH(θ)	$< 10^{-15}$	$2 - \sqrt{3} = 0.26795$	EXACTLY ZERO
HH edge kinetic k_HH(θ)	$< 10^{-15}$	1.00000	EXACTLY ZERO
Total kinetic	$< 10^{-15}$	$3 - \sqrt{3} = \lambda_{E_g}$	EXACTLY λ_{E_g}

The remarkable split $k_{SH}/k_{HH} = (2-\sqrt{3})/1 \approx 0.268$ reflects the √3 structure of the tetrahedral Z-sector geometry, with $k_{SH} = \lambda_{E_g} - 1 = 2 - \sqrt{3}$. Both components are exactly θ-independent, confirming Theorem 3.1 at the edge-resolved level. [STATUS: VERIFIED to machine precision, 3601 sample points.]

3.3 Consequence: Inter-Sector Origin of U(1) Breaking

Corollary 3.2. The $U(1) \to C_3$ symmetry breaking cannot arise from any single-polyhedron quantity. It arises exclusively from the inter-sector O_h/I_h frame mismatch.

Proof: Theorem 3.1 applies independently to both the truncated octahedron (O_h) and the truncated icosahedron (I_h). The key structural observation is that I_h has no 2-dimensional irreducible representation: the I_h irrep dimensions are {1, 3, 3, 4, 4, 5, 5}. Therefore, the 2D E_g eigenspace of O_h cannot be mapped bijectively onto any single I_h irrep. The Z-sector mediator, which lives in this 2D space, must project onto a subspace of a higher-dimensional I_h irrep, introducing the frame mismatch angle $\alpha = \pi/10 = 18°$ (ZS-S6 §3.3, DERIVED).

This mismatch generates the non-abelian holonomy (ZS-S6 Theorem 1, PROVEN) with $||H_{fwd} - H_{bwd}|| = 0.0331$ and CP-violating phase $\Phi_{CP} = 19.06°$. The C_3 subgroup of O_h (rotation by $2\pi/3$ around the [111] axis) acts on the E_g eigenspace as a 120° rotation, providing the 3-fold harmonic structure that distinguishes the Reuleaux triangle from a circle. The C_3 symmetry is inherited from the Z-sector's tetrahedral geometry (n = 3 in the polygon-tetration family, The Book §4.4). [STATUS: DERIVED from Theorem 3.1 + ZS-S6 Theorem 1]

§4. The Variational Chain: Action → Area Minimization

4.1 Z-Sector 1-Loop Effective Action

The Z-Spin action (ZS-F1 §3.1) in the Jordan frame is $S[g, \Phi] = \int d^4x\sqrt{(-g)}[\frac{1}{2}M^2_P(1+A|\Phi|^2)R - \frac{1}{2}M^2_P|\partial\Phi|^2 - V(\Phi)] + S_m$, where $V(\Phi) = (\lambda/4)M^4_P(|\Phi|^2 - 1)^2$. The Z-sector field $\Phi = \rho e^{i\theta}$ is frozen at $\rho = 1$ with mass $m_\varepsilon \approx 1.34\ M_P$ and Compton wavelength $\lambda_C \approx 6\times10^{-35}$ m, which is 61 orders of magnitude shorter than any cosmological scale (ZS-F3). Consequently, the ε-field decouples from perturbation dynamics on all observable scales.

At the frozen attractor $\rho = 1$, the tree-level potential $V = (\lambda/4)M^4_P(1-1)^2 = 0$ is exactly flat in the angular direction θ. This is the Goldstone theorem for $U(1)_Z$: the phase mode is exactly massless at tree level. The cross-sectional shape is therefore determined entirely by quantum corrections—specifically, by the 1-loop effective action:

$$\Gamma_Z[\Omega] = \frac{1}{2} \ln \det(\Delta_\Omega) \qquad (6)$$

where Δ_Ω is the Laplacian on the 2D cross-sectional domain Ω, with boundary conditions inherited from the Gibbons–Hawking–York term $S_{GHY}^{ZS} = [(1+A)/(16\pi G_*)] \int_{\partial V} K\sqrt{h}\, d^3x$ (ZS-Q6 §3.2, DERIVED-under-Regge). The prefactor (1+A) reflects the non-minimal coupling evaluated at the attractor.

4.2 Seeley–DeWitt Expansion

The heat kernel of Δ_Ω on a 2D domain with piecewise-smooth boundary has the asymptotic expansion (Gilkey 1995, McKean–Singer 1967):

$$\mathrm{Tr}(e^{-t\Delta}) = a_0\, t^{-1} + a_{1/2}\, t^{-1/2} + a_1 + O(t^{1/2}) \quad \text{as } t \to 0^+ \tag{7}$$

For a domain Ω with perimeter L and interior angles α_i at corners:

$a_0 = \mathrm{Area}(\Omega)/(4\pi)$ [Weyl term, shape-dependent]

$a_{1/2} = -L/(4\sqrt{\pi})$ [perimeter term, FIXED by Barbier for CW curves]

$a_1 = (1/6)\chi(\Omega) + \Sigma_i\, (\pi/\alpha_i - \alpha_i/\pi)/24$ [Euler characteristic + corners]

The spectral zeta function $\zeta_\Omega(s) = \Sigma_n \lambda_n^{-s}$ relates to the heat kernel via Mellin transform: $\zeta(s) = [1/\Gamma(s)] \int_0^\infty t^{s-1}\, \mathrm{Tr}(e^{-t\Delta})\, dt$. The effective action is $\Gamma_Z = -\frac{1}{2}\zeta'_\Omega(0)$. The a_0 term contributes to the pole at s = 1 of the zeta function. Through the standard regularization procedure, the area-dependent part of $\zeta'(0)$ is determined:

$$\partial\zeta'(0)/\partial(\mathrm{Area}) = -(1/(4\pi))(1 + \gamma_E + \ln \Lambda^2) \tag{8}$$

where $\gamma_E = 0.5772...$ is the Euler–Mascheroni constant and Λ is the UV cutoff, set by the polyhedral lattice scale $m_\varepsilon \approx 1.34\, M_P$ (ZS-F3). Since $\Gamma_Z = -\frac{1}{2}\zeta'(0)$, the sign is:

$$\partial\Gamma_Z/\partial(\mathrm{Area}) = +(1/(8\pi))(1 + \gamma_E + \ln \Lambda^2) > 0 \tag{9}$$

The right-hand side is positive for all $\Lambda > e^{-(1+\gamma_E)/2} \approx 0.45$, which is trivially satisfied for $\Lambda \sim M_P$. [STATUS: DERIVED from standard Seeley–DeWitt expansion + zeta regularization (Gilkey 1995, Vassilevich 2003).]

4.3 The Sign Theorem

Theorem 4.1 (Action-Area Monotonicity). For the Z-sector 1-loop effective action on a 2D domain Ω with UV cutoff Λ at the polyhedral lattice scale: $\partial\Gamma_Z/\partial(\mathrm{Area}) > 0$. Increasing the cross-sectional area increases the effective action. Minimizing Γ_Z requires minimizing the area of Ω. [STATUS: DERIVED]

Remark: This sign is physically natural. A larger cross-section has more degrees of freedom (more eigenmodes of the Laplacian below any given cutoff), increasing the 1-loop partition function and hence the effective action. The Weyl law $N(\lambda) \sim (\mathrm{Area}/(4\pi))\lambda$ (first term) makes this explicit: area controls the spectral density.

4.4 The Blaschke–Lebesgue Isomorphism

The variational problem for the Z-sector cross-section is now precisely defined:

Minimize: $\Gamma_Z[\Omega] = C_1 \cdot \mathrm{Area}(\Omega) + C_2 \cdot L(\Omega) + C_3 \cdot (\text{corner terms}) + \ldots$

Subject to: (C1) $h(\theta) + h(\theta+\pi) = w$ (constant width, from U(1)_Z J-compatibility); (C2) $h + h'' \geq 0$ (convexity).

Since $C_1 > 0$ (Theorem 4.1), C_2 is fixed by Barbier's theorem (no shape-dependence), and $C_3 > 0$ (corner deficits reduce the action), the minimizer of Γ_Z is the same as the minimizer of Area under (C1)+(C2). This is precisely the Blaschke–Lebesgue problem, whose unique solution is the Reuleaux triangle (Theorem 2.2).

Theorem 4.2 (Blaschke–Lebesgue–Z-Spin Isomorphism). The variational problem min Γ_Z under the constant-width constraint (from U(1)_Z) and convexity is isomorphic to the Blaschke–Lebesgue minimum-area problem. The unique solution is the Reuleaux triangle. [STATUS: DERIVED]

4.5 Complete Derivation Chain

Step	Content	Source	Status
1	dim(Z) = 2 → plane curve	ZS-F5 v1.0	PROVEN
2	U(1)_Z → constant width	ZS-F1 + ZS-M7 Thm 4	PROVEN
3	ε frozen → 1-loop Γ_Z	ZS-F3 v1.0	DERIVED
4	∂Γ/∂(Area) > 0	Thm 4.1 (this paper)	DERIVED
5	Convexity h + h″ ≥ 0	Physical requirement	PROVEN
6	BL → Reuleaux triangle	Blaschke–Lebesgue 1915	PROVEN
7	C_3 from tetrahedron	Face-Polygon Corr.	PROVEN

Remaining gap: Step 4 uses the sign of ∂Γ/∂(Area) from the standard Seeley–DeWitt expansion. Explicit verification under the Z–Spin GHY boundary conditions (non–minimal coupling prefactor (1+A)) would close this gap. The sign depends only on the positivity of $(1+\gamma_E+\ln\Lambda^2)$, which is satisfied for any physical UV cutoff. [STATUS: DERIVED, with gap classified as Gate FF7–1.]

§5. The Amplitude 1/8 = 1/(n² − 1)

5.1 Derivation from Convexity Saturation

The non-circularity amplitude is determined by a three-step chain from locked inputs:

Step (i): The Z–sector polygon number is n = 3 (Face–Polygon Correspondence, The Book §4.5, PROVEN). In the polygon–tetration family $z^*(n) = -W_0(-2\pi i/n)/(2\pi i/n)$, n = 3 (triangle) is the unique unstable fixed point with $|f'(z^*)| = 1.0330 > 1$, while n = 4 (square, X–sector) is the first

stable polygon. The critical transition occurs at $n_c = 3.2036$. The Z-sector sits precisely at this instability boundary.

Step (ii): The convexity constraint $h(\theta) + h''(\theta) \geq 0$, applied to $h = w/2 + a_n \cos n\theta$, gives:

$$w/2 + (1 - n^2)\, a_n \cos n\theta \geq 0 \quad \Rightarrow \quad |a_n| \leq w / [2(n^2 - 1)] \qquad (10)$$

For $n = 3$: $|a_3| \leq w/16$.

Step (iii): The area-minimization principle (Theorem 4.1) requires maximizing $(n^2-1)a_3^2$ in Eq. (3), which means saturating the convexity bound: $a_3 = w/16$. The relative amplitude is:

$$a_3/(w/2) = 1/(n^2 - 1)|_{n=3} = 1/8 \qquad (11)$$

This generalizes to other polygon numbers: $n = 5$ gives 1/24, $n = 7$ gives 1/48. Only $n = 3$ is selected by the Z-Spin framework. [STATUS: PROVEN. Each step uses standard convex geometry + PROVEN input $n = 3$.]

5.2 Perturbative No-Go Theorem

Theorem 5.1 (C_3-Forbidden cos(3θ)). The C_3-summed Coleman–Weinberg effective potential $V_{CW}(\theta) = \Sigma_{k=0}^{2} f(\theta - 2\pi k/3)$, where $f(\theta)$ is any polynomial in $\cos^2\theta$, contains no $\cos(3\theta)$ harmonic at any finite order. Only $\cos(6m\theta)$ harmonics ($m = 1, 2, \ldots$) survive the C_3 projection. [STATUS: PROVEN]

Proof: (1) $\cos^{2n}\theta$ expands via the binomial theorem into harmonics $\cos(2k\theta)$ for $k = 0, 1, \ldots, n$. Only even-order harmonics appear. (2) The C_3 average annihilates $\cos(2k\theta)$ unless $2k \equiv 0 \pmod 3$, i.e., $k \equiv 0 \pmod 3$. (3) Surviving harmonics: $\cos(6m\theta)$ with $m = 0, 1, 2, \ldots$. (4) $\cos(3\theta)$ would require $k = 3/2$, which is not an integer. Therefore $\cos(3\theta)$ is absent at all orders. □

5.3 Numerical Verification

The C_3-summed effective potential was computed numerically with frame mismatch angle $\alpha = \pi/10$:

Order	V_3 (cos 3θ amplitude)	V_6 (cos 6θ amplitude)	V_3/V_0
Quartic (4th)	0 (exact, analytical)	0	0
Sextic (6th)	$< 10^{-17}$	0.4956 (relative)	0
Cross-term 4×4	$< 10^{-17}$	0.4729 (relative)	0

At quartic order, the analytical result $V(\theta) = (3/4)(1 + \cos(2\alpha)/2) = 1.0534$ is exactly θ-independent (verified to machine precision over 3,601 sample points). The sextic order produces only $\cos(6\theta)$ with no $\cos(3\theta)$, confirming the no-go theorem numerically. [STATUS: VERIFIED]

Physical resolution: The $\cos(3\theta)$ content of the Reuleaux cross-section is not a perturbative loop correction but a geometric extremum—the solution to a constrained optimization problem (Blaschke–Lebesgue). The 1-loop action provides the optimization criterion (minimize area); the shape, including its Fourier content, is the mathematical answer to that criterion. The perturbative no-go

confirms that the Reuleaux geometry is fundamentally non-perturbative in character, consistent with its origin in a variational principle rather than a loop expansion.

§6. J-Symmetry and Constant-Width Equivalence

6.1 J-Decomposition of the Support Function

The seam involution J (ZS-F5, ZS-M7) acts on the Z-sector field space as $\theta \to \theta + \pi$. Under this action, the support function decomposes into J-symmetric and J-antisymmetric parts:

$$h_+(\theta) = [h(\theta) + h(\theta+\pi)]/2 = w/2 \quad \text{(J-symmetric, constant)}$$
$$h_-(\theta) = [h(\theta) - h(\theta+\pi)]/2 = (w/16)\cos 3\theta \quad \text{(J-antisymmetric)}$$

Verification: $\cos 3(\theta+\pi) = \cos(3\theta + 3\pi) = \cos 3\theta \cdot \cos 3\pi = -\cos 3\theta$, confirming J-antisymmetry of the $\cos 3\theta$ term. The constant-width condition $h(\theta) + h(\theta+\pi) = w$ is equivalent to $h_+ = w/2 = \text{const}$, which is equivalent to the J-symmetric part being direction-independent. The non-circularity $\delta h = h_-$ resides entirely in the J-antisymmetric sector. [STATUS: PROVEN]

Z-sector slot verification: In the Q = 11 transfer operator (ZS-M4), the Z-sector slots are {4, 6}. The phase matrix W_p gives: $\varphi(j=4) = -2\pi/p$ and $\varphi(j=6) = +2\pi/p$. The J-symmetric part $(\varphi_4+\varphi_6)/2 = 0$ for ALL primes (exact, algebraic). The J-antisymmetric part $(\varphi_6-\varphi_4)/2 = 2\pi/p$. The Z-sector phase is PURELY J-antisymmetric, exactly matching the Reuleaux non-circularity structure. [STATUS: VERIFIED]

6.2 Three Levels of Vanishing at σ = 1/2

Level 1 (non-circularity): $\delta h(\theta)$ does NOT vanish at $\sigma = 1/2$. The Reuleaux triangle is not a circle. $\delta h = (w/16)\cos 3\theta \neq 0$ regardless of σ.

Level 2 (J-symmetry violation): ε_J DOES vanish exactly at $\sigma = 1/2$ (ZS-M7 Theorem 4, PROVEN). Geometrically: the constant-width condition holds, meaning the 'width' (J-symmetric part) is direction-independent. Off critical line: $\varepsilon_J(\sigma) = O(|\sigma - 1/2|)$ with slope ≈ 6.10 (ZS-M7 Theorem 4(iii)).

Level 3 (functional equation): $D_\xi(s) = D_\xi(1-s)$ is PRESERVED because: (a) the area change $(\pi-\sqrt{3})w^2/2$ vs $\pi w^2/4$ is a multiplicative constant in B(s); (b) the perimeter is unchanged (Barbier); (c) the corner contribution $\Delta a_1 = 1/3$ is an additive constant in log B(s). All three preserve $s \leftrightarrow 1-s$ symmetry. [STATUS: DERIVED]

Key insight: A constant-width curve is the unique geometry that can be non-circular (have non-trivial δh) while preserving J-compatibility ($\varepsilon_J = 0$). The Reuleaux triangle is the most non-circular such curve (minimum area). It represents the maximum geometric information that can be encoded in the Z-sector cross-section without breaking the functional equation.

§7. Seeley–DeWitt Coefficients: $a_1 = 1/2$

7.1 Coefficient Comparison: Circle vs. Reuleaux

Coefficient	Circle (r = w/2)	Reuleaux (width w)	Difference	s↔1−s
a_0 (area)	$\pi w^2/(16\pi) = w^2/16$	$(\pi-\sqrt{3})w^2/(8\pi)$	factor 0.8973	PRESERVED
a_{1/2} (perim.)	$-\pi w/(4\sqrt{\pi})$	$-\pi w/(4\sqrt{\pi})$	0 (Barbier)	PRESERVED
a_1 (corners)	1/6 = 0.1667	1/6 + 1/3 = 1/2	Δ = 1/3	PRESERVED

7.2 Corner Contribution: Exact Calculation

For a domain with piecewise-smooth boundary and interior angles α_i at the corners, the standard Seeley–DeWitt corner correction (McKean–Singer 1967, Kac 1966) is:

$$\Delta a_1 = \Sigma_i\ (\pi/\alpha_i - \alpha_i/\pi)/24 \qquad (12)$$

For the Reuleaux triangle with three vertices at interior angle $\alpha = \pi/3$:

Per vertex: $(\pi/(\pi/3) - (\pi/3)/\pi)/24 = (3 - 1/3)/24 = (8/3)/24 = 8/72 = 1/9$
Total (3 vertices): $3 \times 1/9 = 1/3$
Grand total: $a_1 = 1/6 + 1/3 = 3/6 = 1/2$

Z-Spin interpretation of the corner contribution:

Corner angle $\pi/3 = \pi/X$ (X-sector dimension imprint)
Per-vertex contribution $1/9 = 1/X^2$
Total contribution $1/3 = 1/X = X/(X \cdot X)$ (dimension ratio)

[STATUS: PROVEN. The corner formula is standard spectral geometry (McKean–Singer). The Z-Spin interpretation (angle = π/X, contribution = 1/X) is DERIVED from the identification X = 3.]

7.3 Functional Equation Preservation

The completion factor B(s) in the spectral determinant D_ξ(s) = B(s) · det(I − L_s) depends on the Seeley–DeWitt coefficients. The three modifications from Reuleaux geometry each preserve the functional equation D_ξ(s) = D_ξ(1−s):

(1) Area reduction (a_0): enters as an overall multiplicative constant in B(s). Since this constant does not depend on s, it factors out of both D_ξ(s) and D_ξ(1−s), preserving the symmetry.

(2) Perimeter (a_{1/2}): UNCHANGED by Barbier's theorem. No modification to B(s).

(3) Corner contribution ($\Delta a_1 = 1/3$): enters as an additive constant in $\zeta(0)$, hence a multiplicative constant $\exp(-1/3)$ in the spectral determinant. This is again s-independent and preserves the functional equation.

[STATUS: DERIVED. The Reuleaux geometry is J-compatible at all Seeley–DeWitt orders.]

§8. Downstream Implications

8.1 Heat Kernel Pipeline: B(s) Closure [OPEN]

The Reuleaux Seeley–DeWitt coefficients provide the first concrete geometric inputs for the ZS-QS §4.3 heat kernel pipeline aimed at deriving the archimedean completion factor B(s). Specifically: $a_0 = (\pi-\sqrt{3})w^2/(8\pi)$ determines the Weyl density (eigenvalue count), $a_{1/2} = -\pi w/(4\sqrt{\pi})$ (unchanged by Barbier) fixes the subleading correction, and $a_1 = 1/2$ constrains the constant term. The heat kernel reconstruction $\mathrm{Tr}(e^{-tH^2}) \approx a_0/\sqrt{t} + a_1 + a_2\sqrt{t} + \ldots$ with these inputs may provide sufficient structure to close the P2 target (ZS-QS §4.1): the identification of B(s) as an archimedean factor. Whether these three coefficients are sufficient, or whether the full eigenvalue distribution of the Reuleaux Laplacian is needed, remains OPEN.

[Dated Update 2026-04-15 — Status Demotion for Cosmological Chain]
The §8.1 Heat Kernel Pipeline was previously identified in ZS-F2 v1.0 §11.8.8(iv) as a BLOCKING gate for the full DERIVED status of the Spectral–Index Projection Theorem (Theorem 11.8) and the F-BMT2 falsification gate. As of 2026-04-15, this blocking role is superseded by two independent lines of work:
First, the Transcendental Budget Lemma (companion work, draft 2026-04-15) provides a direct, structurally bounded derivation of $\varepsilon_higher = \Omega_m(face)\cdot Q^2 - [\eta_topo\cdot Q^2 - \Delta a_2/e - ind^-(D_Z)]$ via the rank-1 β_0-selected Block-Laplacian structure of ZS-M6, without invoking the Reuleaux heat kernel eigenvalue distribution. The lemma bounds $|\varepsilon_higher|$ with explicit margin (4.55% of the F-BMT2 budget), and shows that ε_higher has zero closed-form rational expression by Gelfond–Schneider applied to the Lambert-W fixed point z^*.
Second, the Dimensional Coupling Norm Theorem (ZS-M6 v1.0 §2.2, dated update 2026-04-15) upgrades Δa_2 from a 3-decimal numerical value to the exact rational $315/4807 = 9A/Q$, via the Register-Total Normalization Theorem. Under this upgrade, the structural identity (☆ 11.8) of ZS-F2 §11.8.3 closes to mpmath-arbitrary precision: $\Omega_m(face)\cdot Q^2 = \eta_topo\cdot Q^2 - \Delta a_2/e - ind^-(D_Z) + \varepsilon_higher$, where $\Delta a_2 = 315/4807$ EXACT, $\varepsilon_higher = 0.0477244614209...$ (50-digit). The F-BMT2 gate is PASS with 4.55% margin, independently of whether the Reuleaux heat kernel pipeline is closed.
Consequence for §8.1: the Heat Kernel Pipeline is demoted from BLOCKING to SUPPLEMENTARY with respect to the cosmological chain (i-tetration → face counting → Ω_m). It retains its ORIGINAL motivation as a path toward the Riemann zeta connection: specifically, the identification of B(s) as an archimedean completion factor in the ZS-QS §4.1 P2 target ($\sigma = 1/2$ resonance, §8.2 of this paper). This motivation is independent of the cosmology derivation and remains OPEN.
[STATUS: SUPPLEMENTARY for ZS-F2 §11.8 chain; OPEN for ZS-QS P2 chain.]

8.2 The σ = 1/2 Structural Resonance [HYPOTHESIS]

The spectral zeta function at s = 0 evaluates to $\zeta_\Omega(0) = a_1 = 1/2$ for the Reuleaux domain. This is the same numerical value as the critical line parameter σ = 1/2 in the Riemann Hypothesis. Three distinct mathematical objects take the value 1/2 simultaneously in the Z-Spin framework: (1) the spectral invariant a_1 of the Reuleaux seam; (2) the J-intertwining locus $\varepsilon_J = 0$ (ZS-M7 Theorem 4); (3) the symmetry axis of the functional equation $\xi(s) = \xi(1-s)$. Whether this triple coincidence has a deeper structural origin—connecting the Z-sector's geometric shape to the location of zeta zeros—requires the P1–P4 closure program (ZS-QS §4, all OPEN). [STATUS: HYPOTHESIS]

8.3 Instanton Considerations [OPEN]

The Perturbative No-Go Theorem (§5.2) establishes that cos(3θ) cannot arise from perturbative corrections. A natural question is whether non-perturbative effects (instantons) could provide an independent mechanism. In the Z-Spin framework, the $U(1)_Z$ vortex cores have topological charge (Z-Anchor theorem, ZS-F1 §5.2, PROVEN via $\pi_1(U(1)) = \mathbb{Z}$). The instanton action for $U(1) \to \mathbb{Z}_3$ breaking would be $S_{inst} \sim S_{tunnel}/3$, where $S_{tunnel} = 5\pi/A \approx 196.1$ (ZS-M3, DERIVED). This gives an exponentially suppressed amplitude $e^{-S_{inst}} \sim e^{-65}$, which is negligible. The instanton mechanism is therefore consistent with (but not needed for) the variational derivation. The Reuleaux cross-section is selected by the Blaschke–Lebesgue principle, not by instanton dynamics. [STATUS: OPEN (instanton analysis complete but classified as supplementary).]

8.4 Product Lattice Holonomy [OPEN]

The Single-Polyhedron U(1) Exactness Theorem (§3) establishes that U(1) breaking arises from the inter-sector coupling. The explicit computation of the holonomy on the full product lattice $\Gamma_X \otimes \Gamma_Y$ (dimension 24×60 = 1440) would provide an independent confirmation of the C_3 harmonic structure. Preliminary analysis shows: the C_3 subgroup of O_h acts on the E_g eigenspace as 120° rotations, the C_5 subgroup of I_h acts on the $\mathbb{Z}_5$ character space as 72° rotations, and their combined action on the 2D Z-sector has period lcm(3,5) = 15. The full product lattice computation is deferred to future work. [STATUS: OPEN]

§9. Falsification Gates and Non-Claims

9.1 Falsification Gates

Gate	Condition	Falsification Criterion	Status

FF7-1	∂Γ_Z/∂(Area) > 0	Sign reversal under ZS GHY boundary conditions	DERIVED
FF7-2	BL uniqueness	Non-Reuleaux CW curve with smaller area found	PROVEN (none)
FF7-3	n = 3 selection	Z−sector polygon number ≠ 3	PROVEN (stable)
FF7-4	C_3 No-Go	cos(3θ) found in perturbative CW potential	PROVEN (impossible)
FF7-5	$a_1 = 1/2$	Corner calculation error found	PROVEN (exact)
FF7-6	Barbier protection	Perimeter varies for CW curves	PROVEN (theorem)
FF7-7	U(1) exactness	θ-dependence on single polyhedron found	PROVEN ($< 10^{-15}$)

Current status: All gates OPEN (no falsification triggered). FF7-1 is the only gate with non-trivial DERIVED status; explicit verification under Z-Spin GHY boundary conditions would upgrade to PROVEN. Gates FF7-2 through FF7-7 are protected by mathematical theorems and cannot be falsified by observation.

9.2 Non-Claims

NC-F7.1: This paper does NOT claim a proof of the Riemann Hypothesis. The structural resonance $a_1 = 1/2 = \sigma_{critical}$ is classified as HYPOTHESIS. The gap between the finite-dimensional Z-Spin operator and the infinite-dimensional ζ(s) (P1–P4 of ZS-QS §4) remains OPEN.

NC-F7.2: This paper does NOT claim the area deficit (2√3−π)/4 ≈ 0.0806 equals A = 35/437 ≈ 0.0801. The 0.67% proximity is OBSERVATION without derivation chain. (2√3−π)/4 is transcendental; 35/437 is rational. They cannot be equal.

NC-F7.3: This paper does NOT claim the Reuleaux cross-section has been experimentally observed. It is a theoretical prediction derived from the Z-Spin action.

NC-F7.4: The amplitude 1/8 is NOT a perturbative coefficient. Theorem 5.1 proves that cos(3θ) is forbidden at all perturbative orders. The value $1/8 = 1/(n^2-1)$ is a geometric constant from convexity saturation.

NC-F7.5: This paper does NOT claim the 50/50 curvature split has direct physical consequences beyond the structural analogy with X/Y mediation. The split is a mathematical fact about the Reuleaux triangle; its physical interpretation is HYPOTHESIS.

§10. Conclusion

We have derived the Reuleaux triangle as the effective cross-sectional geometry of the Z-sector mediation boundary through a seven-step chain from the Z-Spin action with zero free parameters. The central result is the Blaschke–Lebesgue–Z-Spin Isomorphism (Theorem 4.2): minimizing the Z-sector 1-loop effective action Γ_Z under the constant-width constraint (from U(1)_Z J-compatibility) is mathematically equivalent to the Blaschke–Lebesgue minimum-area problem, whose unique solution is the Reuleaux triangle.

Three new theorems are established: the Single-Polyhedron U(1) Exactness Theorem (Theorem 3.1, PROVEN), establishing that U(1) breaking is exclusively inter-sectoral; the Action-Area Monotonicity Theorem (Theorem 4.1, DERIVED), providing the sign that connects action minimization to area minimization; and the Perturbative No-Go Theorem (Theorem 5.1, PROVEN), confirming that the Reuleaux geometry is non-perturbative. The non-circularity amplitude $1/8 = 1/(n^2-1)$ at $n = 3$ is a geometric extremum from convexity saturation. Three Seeley-DeWitt coefficients characterize the cross-section, with the notable result $a_1 = 1/2$ from the corner contribution $\Delta a_1 = 1/X = 1/3$.

These results complete the Z-sector's foundational characterization within the F-series, adding geometric cross-section to its algebraic (dim = 2), dynamical ($L_{XY} \equiv 0$), and polyhedral (tetrahedron pair) properties. The Reuleaux geometry provides concrete spectral inputs for the Berry-Keating heat kernel pipeline and establishes a structural link between the Z-sector's shape and the critical line $\sigma = 1/2$, pending resolution of the P1-P4 closure program.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/mpmath, 50-digit precision) is publicly available.

Verification script: ZS_F7_verify_v1_0.py. Dependencies: Python 3.10+, NumPy, SciPy, mpmath. Execution: python3 ZS_F7_verify_v1_0.py. Expected output: 18/18 PASS, exit code 0. Covers: TO construction (3 tests), E_g eigenspace (3), θ-independence (4), Reuleaux geometry (3), perturbative no-go (2), Seeley–DeWitt (2), anti-numerology (1). No external data files required.

Appendix A. Complete Derivation Chain

#	Statement	Source	Status
1	$A = 35/437 = \delta_X \cdot \delta_Y$	ZS-F2 v1.0	LOCKED
2	(Z,X,Y)=(2,3,6), Q=11	ZS-F5 v1.0	PROVEN
3	$U(1)_Z$: $\Phi \to e^{i\alpha}\Phi$	ZS-F1 v1.0 §3.2	PROVEN
4	$L_{XY} \equiv 0$ (Z-mediation)	ZS-F1 v1.0	PROVEN
5	n=3 (Face-Polygon)	The Book §4.5	PROVEN
6	$\varepsilon_J=0 \Leftrightarrow \sigma=1/2$	ZS-M7 Thm 4	PROVEN

7	ε frozen, m_ε≈1.34M_P	ZS-F3 v1.0	DERIVED
8	U(1) exact, single poly.	Thm 3.1 (this paper)	PROVEN
9	∂Γ_Z/∂(Area)>0	Thm 4.1 (this paper)	DERIVED
10	BL: min area CW=Reuleaux	Blaschke–Lebesgue	PROVEN
11	$a_3/(w/2)=1/(n^2-1)=1/8$	Convexity saturation	PROVEN
12	Perturbative No-Go	Thm 5.1 (this paper)	PROVEN
13	$\Delta a_1=1/X=1/3$	Seeley–DeWitt corners	PROVEN
14	$a_1=1/6+1/3=1/2$	Steps 13+standard	PROVEN

Appendix B. Cross-Reference Table

Paper	Content	Direction	Relation
ZS-F1 v1.0	Action, U(1)_Z, L_XY=0	Input→ZS-F7	LOCKED
ZS-F2 v1.0	A=35/437, polyhedra	Input→ZS-F7	LOCKED
ZS-F3 v1.0	ε-field freezing, m_ε	Input→ZS-F7 §4	DERIVED
ZS-F5 v1.0	dim(Z)=2, Q=11, J	Input→ZS-F7 §1,§3	PROVEN
ZS-M3 v1.0	Regge holonomy δφ=A	Input→ZS-F7 §8	DERIVED
ZS-M6 v1.0	Heat kernel, BL verify	Input→ZS-F7 §4	VERIFIED
ZS-M7 v1.0	J-involution, ε_J=0	Input→ZS-F7 §6	PROVEN
ZS-S6 v1.0	O_h/I_h mismatch	Input→ZS-F7 §3.3	PROVEN
ZS-M4 v1.0	Spectral bridge, BK	ZS-F7→M4 (a_1)	DOWNSTREAM
ZS-QS v1.0	IRE, heat kernel pipe	ZS-F7→QS (SD)	DOWNSTREAM
ZS-S4 v1.0	Higgs VEV, CW mech	ZS-F7→S4	DOWNSTREAM
ZS-Q6 v1.0	Kelvin cell, boundary	ZS-F7→Q6	DOWNSTREAM

Appendix C. Anti-Numerology Analysis

C.1 Area Deficit Proximity

The normalized area deficit $(2\sqrt{3}-\pi)/4 = 0.08063$ is 0.67% from $A = 35/437 = 0.08009$. Monte Carlo: among 500,000 random expressions of the form $(a\sqrt{b} - c\pi)/d$ with a,b,c,d integers in [1,10], approximately 0.8% fall within 0.67% of A. Furthermore, $(2\sqrt{3}-\pi)/4$ is transcendental while 35/437 is rational; they cannot be algebraically related. Status: OBSERVATION. NON-CLAIM (NC-F7.2).

C.2 Amplitude 1/8

The ratio 1/8 = 0.125 is DERIVED, not observed: it equals $1/(n^2-1)$ at the PROVEN input n = 3. The derivation chain (n = 3 → convexity bound $w/(2(n^2-1))$ → saturation → $a_3 = w/16$ → ratio = 1/8) is complete with no free parameters. No Monte Carlo scan is needed. Status: PROVEN.

C.3 Corner Contribution 1/3

The value 1/3 = 1/X arises from exactly three vertices at angle π/3, giving Σ(3−1/3)/24 = 1/3. This is exact algebra from the Seeley–DeWitt corner formula. Among the 13 Archimedean solids, only the truncated tetrahedron shares the vertex angle π/3. The coincidence 1/3 = 1/X is structural (the Z-sector tetrahedron has triangular faces with angle π/3 = π/X). Status: PROVEN.

C.4 Seeley–DeWitt a_1 = 1/2

The value a_1 = 1/2 coincides with σ = 1/2 (Riemann critical line). This is classified as HYPOTHESIS (NC-F7.1) because: (i) 1/2 is the most common rational number in mathematics, appearing in hundreds of contexts; (ii) no derivation chain connects a_1 to σ_{critical}; (iii) the P1–P4 gap remains. The coincidence is noted for future investigation but does NOT constitute evidence for RH.

Appendix D. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] TO Construction	3	3/0	V=24, E=36, F=14, valence=3
[B] E_g Eigenspace	3	3/0	λ=3−√3, deg=2, orthonormality < 10^{-16}
[C] θ-Independence	4	4/0	Vertex, SH, HH, total: all < 10^{-15}
[D] Reuleaux Geometry	3	3/0	Area=(π−√3)/2, L=πw, 1/8 identity
[E] Perturbative No-Go	2	2/0	Quartic V_3=0 (exact), Sextic V_3=0 (<10^{-17})
[F] Seeley–DeWitt	2	2/0	Corner = 1/9 per vertex, a_1 = 1/2
[G] Anti-Numerology	1	1/0	MC: P(area match) < 0.8%

TOTAL: 18/18 PASS — 100% pass rate. All tests are computationally verified (not structurally asserted). Script exits with code 0 (all pass) or 1 (any fail). Dependencies: Python 3.10+, NumPy, SciPy, mpmath. No external data files required.

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v1.0.) Established: seven-step variational derivation chain (Z-Spin action → Reuleaux triangle), Single-Polyhedron U(1) Exactness Theorem (Thm 3.1), Action-Area Monotonicity Theorem (Thm 4.1), Blaschke–Lebesgue–Z-Spin Isomorphism (Thm 4.2), Perturbative No-Go Theorem (Thm 5.1), Seeley–DeWitt corner analysis ($a_1 = 1/2$), J-symmetry constant-width equivalence. Three new results negative: area deficit ≠ A (NC-F7.2), perturbative cos(3θ) forbidden (Thm 5.1), instanton amplitude negligible (§8.3). Verification: 18/18 PASS.

[Dated Update 2026-04-15 — Version History Entry]
[Dated Update 2026-04-15]: §8.1 status demoted from BLOCKING to SUPPLEMENTARY for the cosmological derivation chain, following two independent closures. First, the Transcendental Budget Lemma (companion work) bounds ε_higher in F-BMT2 with 4.55% margin via direct Block-Laplacian argument. Second, ZS-M6 v1.0 §2.2 (dated update 2026-04-15) upgrades Δa_2 to exact rational 9A/Q = 315/4807 via the Dimensional Coupling Norm Theorem, eliminating the 4-decimal precision bottleneck that the Heat Kernel Pipeline was originally intended to close. The §8.1 Pipeline retains its original motivation as a path to the Riemann zeta connection (§8.2 σ = 1/2 resonance, ZS-QS §4.1 P2 target), which is structurally independent of the cosmology chain and remains OPEN. No §8.1 content is deleted: the Seeley–DeWitt coefficient inputs (a_0, a_{1/2}, $a_1 = 1/2$) and the OPEN status for P2 closure are preserved as stated. The three Reuleaux coefficients from §7 are unchanged and remain the primary concrete geometric inputs for any future Riemann zeta work. No prior content deleted; v1.0 label maintained; 18/18 PASS unchanged.